Internship Report

Federated Gradient-Refined Adaptive Averaging for Intrusion Detection:

Faithful Reproduction, Convergence Defect Diagnosis, and a Privacy-Compliant Extension (FedGRAA-v2)

A report submitted in partial fulfilment of the requirements for the research internship

submitted by

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Certificate

This is to certify that the work contained in this internship report entitled **"Federated Gradient-Refined Adaptive Averaging for Intrusion Detection: Faithful Reproduction, Convergence Defect Diagnosis, and a Privacy-Compliant Extension (FedGRAA-v2)"**, submitted by **Nikhil Pravin Pise** (Roll No.: ⟨*Roll No.*⟩), ⟨*Department*⟩, ⟨*Home Institution*⟩, is a bonafide record of the research internship carried out by him under the supervision of ⟨*Professor Name*⟩, ⟨*Department*⟩, Indian Institute of Technology Guwahati, and the day-to-day mentorship of ⟨*Mentor Name*⟩, during the period ⟨*Start Date*⟩ to ⟨*End Date*⟩.

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Parts of the literature triage, drafting, and code development in this internship used AI-assistant tooling under my direction; every reported number was produced by, and independently re-verified against, the compute artifacts in the accompanying repository.

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Abstract

This report documents a research internship on federated-learning intrusion detection: faithfully reproduce the FedGRAA adaptive-aggregation method, explain any divergence from its reported results, correct its convergence theory, and design a privacy-compliant extension. Reproducing the printed algorithm exactly, I measured 0.00% rare-class accuracy on CIC-IDS-2017 against the manuscript's claimed 88.88% (33.33% reproduced on UNSW-NB15 against 21.09% claimed). A mechanistic diagnosis traced the divergence to a missing simplex-projection step: the learned aggregation weight leaves the probability simplex in round 1 in all 120 unprojected cells, turning the aggregation into unbounded affine extrapolation and violating the paper's own convergence lemma. A minimal two-part correction recovers 89.3%/69.2% rare-class accuracy, and the guarantee was restated as gradient-mapping stationarity $O(1/T) + O(\zeta^2)$ under a Polyak-Łojasiewicz condition. The extension, FedGRAA-v2, removes the design's two privacy violations through a secure-aggregation protocol reconstructing the exact composite-loss gradient, reaching 93.47%/69.66% rare-class accuracy across 960 cells at 6.00×-to-3.00× communication cost, with 111/112 paired significance tests rejecting the null. An adversarial evaluation shows the same up-weighting mechanism is exploitable and that successfully defended cells collapse rare-class recall to near 0% - a trade-off none of the evaluated defenses avoids.

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Nomenclature

Roman capitals

B	client mini-batch size (64)
D_k, $\lvert D_k \rvert$	client k's local dataset and its size
D_V; $D_V^{h,m,n}$	held-out validation data; pooled set and head/median/rare-class strata
E, E'	local training epochs (2); weight-refinement inner epochs (5)
$G_{\eta'}(w)$	projected gradient mapping at step size η'
K	number of participating clients (14 CIC-IDS-2017; 9 UNSW-NB15)
L_k	client k's local loss
L_h, L_m, L_u	validation losses on the head, median, and rare-class strata
L_a	common-class loss, the lower of the two common strata, $\min(L_h, L_m)$
L'	composite objective $(1-\gamma)L_a + \gamma L_u$
P, P_k	flattened model parameters; client k's contribution
S_t	client set selected at round t
T	total communication rounds (200)

Roman lowercase

c_k	client k's count-weighted contribution to the composite gradient
d	Cohen's effect size (paired Wilcoxon comparisons); model input width
f	tolerable client dropout/collusion fraction ($f < K/3$)
m	gradient-refresh interval for the federated backward pass
n	number of protected rare classes (3)
n_{arm}, n_n	SecAgg-pooled counts of the common and rare strata
$n_{\text{arm},k}$, $n_{n,k}$	client k's own counts in those strata
t	SecAgg reconstruction threshold (default $\lfloor K/2 \rfloor + 1$)
w, w_k	learned aggregation-weight vector; client k's weight

Greek letters

β_{eff}	effective smoothness constant of the composite objective
γ	personalization coefficient trading common-class vs rare-class loss
Δ	probability simplex $\{w \in \mathbb{R}^K : w_k \geq 0, \sum_k w_k = 1\}$
ε, δ	differential-privacy budget parameters
ζ, $\tilde{\zeta}$	inexactness bound of the gradient oracle; its measured value
η, η'	client learning rate (0.01); weight learning rate (0.1 literal, 0.01 corrected)
η_{fake}	attacker fake-update step size in the signature attack
θ_g, θ_k	global model parameters; client k's local parameters
θ_g'	candidate aggregate broadcast for the federated backward pass
Π_Δ	Euclidean projection onto Δ (Duchi et al.)
σ	per-seed standard deviation of an adversarial metric

Acronyms

ASR	attack success rate
CI	confidence interval
DP	differential privacy
FedGRAA	Federated Gradient-Refined Adaptive Averaging
FL	federated learning
IDS, NIDS	(network) intrusion detection system
IID, non-IID	(non-)independent and identically distributed
MLP	multilayer perceptron
PL	Polyak-Łojasiewicz (condition)
SecAgg	secure aggregation

Chapter 1

Introduction

1.1 Motivation

An intrusion detection system (IDS) monitors network traffic to detect and help prevent attacks. Traditional IDS deployments depend on *centralized* data collection: traffic from every monitored network is pooled at one site and a detection model is trained on the aggregate. That centralization is the deployment blocker. Raw network telemetry is among the most privacy-sensitive data an organization holds; pooling it across organizational boundaries is frequently prohibited, and it concentrates breach risk at a single point. Federated learning (FL) removes the pooling requirement entirely: each participating site trains a local model on its own traffic and shares only model parameters with a coordinating server, which aggregates them into a global model [13, 20].

FL's own aggregation rule, however, creates a second problem specific to intrusion detection. The standard aggregator, FedAvg, weights each client's contribution by its local dataset size alone, under the assumption that client data is roughly homogeneous and balanced. Real network telemetry is not: some clients see frequent, common intrusions such as port scans and denial of service, while the attack classes that matter most operationally - infiltration, SQL injection, Heartbleed-class exploits - are exactly the ones seen rarely, by few clients, in small quantity. Under size-proportional weighting, a client holding the only handful of examples of a rare attack class has negligible influence on the aggregate, however informative its gradient is for that class. The global model's ability to detect the rarest, highest-consequence attacks degrades accordingly. This is a structural under-representation problem, not a training-instability problem.

FedGRAA [1] targets this directly: it replaces the fixed size-proportional weight with a learned, γ-tunable aggregation weight that is re-optimized every round against a composite validation objective explicitly separating common-class and rare-class loss.

1.2 The internship project

The internship reproduced the published FedGRAA method - *"Federated Gradient-Refined Adaptive Averaging for Intrusion Detection Systems"* [1], a manuscript under review at the time of writing - from scratch in TensorFlow/Keras, measured its claims honestly, diagnosed the divergences found, corrected its convergence theory, and designed a novel privacy-compliant extension. The work proceeded in three stages:

Stage A - faithful reproduction. Re-implement the printed algorithm exactly, including its stated datasets, architecture, and hyperparameters, with every experiment cell persisted to a provenance-embedded artifact.

Stage B - critique, corrected theory, and ablation. Root-cause the reproduction's divergences mechanistically, restate the convergence guarantee correctly, and isolate which design element is responsible for which failure via a controlled ablation.

Stage C - extension. Remove the design's two structural privacy violations in a privacy-compliant redesign (FedGRAA-v2), and evaluate it against the full baseline frontier including adversarial robustness.

The internship was carried out under the supervision of an IIT Guwahati professor with day-to-day PhD-student mentorship. All implementation and experimental artifacts are in the accompanying repository.

1.3 Objectives

The work was scoped to five objectives, each of which maps to a results section of Chapter 4:

O1. Faithful reproduction with persisted artifacts - re-implement the published algorithm exactly and reproduce its reported headline results, or measure the gap precisely (Section 4.2).

O2. Root-cause any divergence mechanistically - not merely observe that reproduction differs from claim, but identify the design element responsible and demonstrate it causally (Section 4.3).

O3. Correct the convergence theory - restate the guarantee the actual (non-convex, constrained) system supports, and validate the minimal algorithmic fix with statistical rigor across seeds (Section 4.4).

O4. Remove the design's privacy violations - the mechanism as published requires the server to already know which classes are rare and to hold a pooled labelled validation set, both in tension with FL's decentralization premise; the extension must eliminate both without changing the objective (Sections 3.6 and 4.5).

O5. Evaluate against the full baseline frontier, including adversarial robustness - 19 methods over 10 seeds with paired significance testing, plus a measured attack-surface analysis of the mechanism itself (Sections 4.5–4.8).

1.4 Summary of contributions

1. **The defect.** A mechanistic diagnosis showing the printed FedGRAA weight update has no mechanism to remain a valid convex combination: the learned weight leaves the probability simplex in round 1 in all 120 unprojected experiment cells, turning aggregation into unbounded affine extrapolation and invalidating the paper's own convergence lemma (Sections 3.3, 4.3).
2. **The fix.** A minimal correction - Euclidean projection onto the simplex plus a tenfold-smaller weight learning rate - that restores rare-class accuracy to 89.3%/69.2% on the two benchmarks, validated across 10 seeds in a 2×2 factorial ablation (Section 4.4).
3. **The corrected theorem.** A restated convergence guarantee: gradient-mapping stationarity $O(1/T) + O(\zeta^2)$ under a Polyak-Łojasiewicz condition, extended to the inexact gradient oracle of the privacy-compliant design (Section 3.5).
4. **FedGRAA-v2.** A privacy-compliant redesign that replaces both oracle assumptions - server knowledge of which classes are rare, and a server-held labelled validation set - with a secure-aggregation two-phase protocol reconstructing the *exact* composite-loss gradient (Section 3.6).
5. **The evaluation, including honest negative results.** A 960-cell campaign (10 seeds on the 830-cell main grid, 5 on the T7/T7G arms) against 19 baselines (93.47%/69.66% best rare-class accuracy; 111 of 112 paired significance tests rejecting the null) plus an adversarial evaluation finding that the up-weighting mechanism is exploitable by construction and that defended cells collapse rare-class recall to near 0% - a trade-off no evaluated defense avoids (Sections 4.5, 4.8).

1.5 Outline of the report

Chapter 2 reviews federated learning, adaptive aggregation, robust aggregation, and poisoning attacks and defenses in federated intrusion detection, and positions the present work. Chapter 3 presents the problem formulation, FedGRAA as published, the diagnosed defect, the minimal correction, the corrected convergence theory, and the FedGRAA-v2 extension. Chapter 4 reports the staged experimental results: reproduction, diagnosis, ablation, the FedGRAA-v2 frontier, significance, overhead, and adversarial robustness. Chapter 5 discusses implications and limitations. Chapter 6 concludes and outlines future work.

Chapter 2

Related Work

2.1 Federated learning and FedAvg

FL trains a global model across decentralized clients without sharing local data. At round t, the server broadcasts θ_g^t to a fixed client set S_t; client k starts from the broadcast θ_g^t and runs local optimization for E epochs on its own data D_k,

$$\theta_k^{(e+1)} \leftarrow \theta_k^{(e)} - \eta \nabla L_k(\theta_k^{(e)}; b), \quad \theta_k^{(0)} = \theta_g^t, \quad e = 0, \ldots, E-1, \tag{2.1}$$

returning $\theta_k^{t+1} = \theta_k^{(E)}$; and the server aggregates by dataset-size-proportional weighting, FedAvg [20]:

$$\theta_g^{t+1} = \sum_{k \in S_t} \frac{|D_k|}{|D|} \theta_k^{t+1} \tag{2.2}$$

Algorithm 1 Federated round (FedAvg; the loop FedGRAA modifies)

1: **for** $t = 1 \ldots T$ **do**
2: server broadcasts θ_g^t to clients $k \in S_t$
3: each client runs E local epochs (Eq. (2.1)), returns θ_k^{t+1}
4: server aggregates $\theta_g^{t+1} \leftarrow \sum_k (|D_k|/|D|)\, \theta_k^{t+1}$ (Eq. (2.2))
5: **end for**

The global objective is the size-weighted average of local losses, $L(\theta) = \sum_k (|D_k|/|D|) L_k(\theta)$. Under Eq. (2.2) a client's influence on θ_g is *exactly* proportional to $|D_k|$. Under non-IID data - in particular when the classes that matter most are held by the smallest-data clients - this systematically dilutes their contribution.

2.2 Adaptive aggregation

Adaptive aggregation weighting has converged on three signal families: alignment with the global update direction (FedAWA [29]), client loss or accuracy (FedNolowe [17], PACT [28]), and statistical outlier filtering (MESAS [16], CrowdGuard [24], FreqFed [10]). Related adaptive rules in the same space include FedAPA [31], DA-PFL [38], FedMADE [30] and XAI-SOH-FL [3]. In imbalanced intrusion detection, all three systematically suppress exactly the clients FedGRAA is designed to protect: divergent, high-loss, minority-class clients holding the rarest attacks look identical, under these signals, to noise or attackers.

2.3 The counter-current

A counter-current to suppressing unusual updates exists. FedAWARE [39] maximizes gradient *diversity* rather than penalizing it. Jiang et al. [12] and AdaAggRL [36] learn weights via reinforcement learning that explicitly protect rare contributions from being filtered out. CELM [33] up-weights clients with discriminative evidence for underrepresented classes via data-free logit-maximization probing, requiring no server-held validation data; SpectralFed [32] and CoAst [37] estimate contribution data-free via gradient spectral entropy and update-sign agreement. Precision-weighted aggregation [23] derives weights from gradient variance. In intrusion detection specifically, FedPGO [4] reweights clients by gradient orthogonality on CIC-IDS-2017 - the same domain, the same dataset, a closely related mechanism family - and NIDS-FGPA [34] weights by gradient similarity for communication efficiency.

2.4 Trust-based robust aggregation

FLTrust [6], RAGA [43], and FedTruth [9] all assume, in one form or another, a server-held reference - a root dataset (FLTrust) or a trusted validation anchor - structurally identical to the D_V dependency the original FedGRAA shares and FedGRAA-v2 removes. FedTruth explicitly calls this assumption impractical; CoAst independently describes a server reference set as "difficult to obtain in practical FL scenarios;" CELM's own abstract restates the concern. This is a known, common, fixable shortcut across the adaptive-aggregation literature, not a defect specific to FedGRAA.

2.5 Classic optimizers and poisoning defenses

FedAvg [20], FedProx [18], FedOpt [22], SCAFFOLD [15] and FedNova [35] are the field-standard, non-personalization-aware optimizers: they correct systems and statistical heterogeneity in the generic sense, but none carries an explicit rare-class objective, and they are evaluated here as context rather than as the primary competitor set.

FL's openness creates a poisoning surface: malicious clients can submit label-flipped or directly model-poisoned updates (the "pill" attack of Guo et al. [11]). Published defenses in the NIDS domain include SecFedNIDS [41], a consensus-filter/SOS defense evaluated on UNSW-NB15, SB-FedNIDS [40], which targets class imbalance directly, and WeiDetect [25], evaluated against a comparable threat model; Ma et al. [19] survey the wider aggregation-defense landscape. None of these - nor any adjacent adaptive-aggregation method above - asks the question this internship's adversarial evaluation asks directly: does an aggregation rule that deliberately up-weights statistically unusual client contributions thereby create an attack surface an adversary can exploit by construction? AdaAggRL [36], whose RL-learned weighting is explicitly framed as adversarially robust, is the closest adjacent design and is reused unmodified as a third defense arm.

2.6 Positioning

FedGRAA sits inside the counter-current of Section 2.3, at a point none of these methods occupies: a single-global-model aggregation rule with an explicit γ-tunable trade-off between rare-class emphasis and conventional size-proportional weighting, an exact (not approximated) gradient against the rare-class objective, and evaluation on rare-class recall specifically - the axis the related methods' own papers do not report. The claim is not that this region is unoccupied - FedPGO occupies an adjacent point on the same dataset - but that this specific point is unreported, and Chapter 4 evaluates it against the closest fourteen competitors plus five classic optimizers directly, rather than asserting the gap.

Chapter 3

Proposed Methodology

This chapter carries the internship's technical core: what the printed algorithm specifies (Section 3.1, 3.2), the defect found in it (Section 3.3), the minimal correction (Section 3.4), the corrected convergence theory (Section 3.5), the FedGRAA-v2 privacy-compliant redesign (Section 3.6), the attack surface the mechanism itself creates (Section 3.7), and the verification discipline the results rest on (Section 3.8).

3.1 Problem formulation

An FL system trains a global model over K clients that never share raw data. Each round t, the server broadcasts θ_g^t; client k updates locally (Eq. (2.1)) and returns θ_k^{t+1}; the server aggregates. FedGRAA replaces Eq. (2.2)'s fixed size-proportional weighting with a learned weight vector w^t constrained - in intent - to the probability simplex

$$\Delta = \{w \in \mathbb{R}^K : w_k \geq 0,\ \textstyle\sum_k w_k = 1\}, \tag{3.1}$$

so that $\theta_g^{t+1} = \sum_k w_k^t\, \theta_k^{t+1}$ is a convex combination.

Each round, the server forms a candidate aggregate $\theta_g' = \sum_k w_k^{t-1}\theta_k^t$ using the previous round's weights, then refines w for E' inner epochs against a composite validation loss that explicitly separates common-class and rare-class performance:

$$L_a = \min\big(L_h(\theta_g'; D_V^h),\ L_m(\theta_g'; D_V^m)\big) \tag{3.2}$$

$$L'(\theta_g') = (1-\gamma)\, L_a(\theta_g'; D_V^h, D_V^m) + \gamma\, L_u(\theta_g'; D_V^n) \tag{3.3}$$

where D_V^h / D_V^m are validation data split at the median class frequency and D_V^n is held-out data for the n rarest classes. The personalization coefficient $\gamma \in [0,1]$ sets the trade-off: $\gamma \to 1$ prioritizes the rare-class loss L_u; $\gamma \to 0$ recovers ordinary size-agnostic behaviour. The n rarest classes are, in the published design, an oracle list the server is assumed to know in advance. The weight gradient follows the chain rule through the aggregate: since $\theta_g' = \sum_k w_k P_k$ with P_k the flattened client parameters,

$$\frac{\partial L'}{\partial w_k} = \big\langle \nabla_P L'(\theta_g'),\ P_k \big\rangle. \tag{3.4}$$

3.2 FedGRAA as published

Algorithm 2 states the weight update *as printed* in the reproduced manuscript, including the Adam optimizer the manuscript's own experimental section specifies for both the client update and the weight update (confirmed by its correction note and by the reference implementation, which labels this line "Adam step on w (eq. 10)").

The initial weights use static reparameterization - inverse-size weighting $w_k^0 \propto 1/|D_k|$ - a fixed control that over-corrects toward small clients without adaptivity:

$$\theta_g^{t+1} = \sum_k \left[\frac{1/|D_k|}{\sum_j 1/|D_j|}\right] \theta_k^{t+1} \tag{3.5}$$

Algorithm 2 FedGRAA weight update - as printed (defective)

procedure FedGRAA(θ_k^t, w^{t-1})
  for $e = 1 \dots E'$ **do**
    $\theta'_g \leftarrow \sum_k w_k^{t-1,e}\, \theta_k^t$
    $L_a \leftarrow \min\big(L_h(\theta'_g; D_V^h),\, L_m(\theta'_g; D_V^m)\big)$
    $L' \leftarrow (1-\gamma)\, L_a + \gamma\, L_u(\theta'_g; D_V^n)$
    $g \leftarrow \nabla_w L'(\theta'_g)$ ▷ Eq. (3.4)
    $w^{t-1,e} \leftarrow$ Adam-step($w^{t-1,e-1}$, g)
  end for
  return $w^t \leftarrow w^{t-1,E'}$ ▷ no projection - see Section 3.3
end procedure

3.3 The diagnosed defect

The headline internship finding is that Algorithm 2 has **no mechanism to remain a valid convex combination**. The Adam update on w is unconstrained: nothing projects w back onto Δ after the inner-epoch refinement. Section 4.3 shows empirically that this is not a cosmetic gap - w leaves the simplex in round 1 in all 120 unprojected experiment cells, at every γ, on both datasets, with zero exceptions. Once $w \notin \Delta$, the "weighted average" $\sum_k w_k \theta_k$ is no longer a convex combination but an unbounded affine extrapolation, and the composite loss it produces can - and does - diverge.

The manuscript's own Appendix A.1 convergence proof cannot cover this behaviour, for three independent reasons:

1. its convexity assumption on L' is false for the paper's own stated tanh-MLP loss surface;
2. the composite term $L_a = \min(L_h, L_m)$ is non-smooth exactly in the region training drives the iterate toward; and
3. its Lemma 2 tacitly assumes $w \in \Delta$ - the very property the printed update does not maintain - so the proof's premise is violated from round 1.

3.4 The minimal correction

The minimal change that restores the intended behaviour is an explicit Euclidean projection onto the simplex after the inner-epoch refinement, plus a tenfold-smaller weight learning rate $\eta' = 0.01$ (the literal 0.1 overshoots the simplex boundary in a single step). With Duchi et al.'s [7] projection:

$$w^t = \Pi_\Delta\big(w^{t-1,E'} - \eta'\, \partial L'(w^{t-1,E'})\big), \tag{3.6}$$

implemented as Algorithm 3; $E' = 5$ is unchanged. This corrected variant was kept strictly diagnostic - it is evaluated against the literal arm in Section 4.4 but never merged into the reproduction baseline, so the reproduction numbers of Section 4.2 remain a faithful measurement of the printed algorithm.

3.5 Corrected convergence theory

The original Theorem 1 assumed L' convex, which is false for the stated tanh-MLP loss surface, and is therefore not repeated here. Stage B restated the guarantee the projected-gradient-mapping framework actually supports for a non-convex, simplex-constrained objective (Proposition 3.1); Stage C extended it to the inexact gradient oracle the privacy-compliant design introduces (Theorem 3.1).

Proposition 3.1 (Feasibility + gradient-mapping stationarity)**.** *Let L' have bounded subgradients on Δ. Projected subgradient descent $w^t = \Pi_\Delta(w^{t-1} - \eta'\, \partial L'(w^{t-1}))$, $\eta' \le 1/\beta_{\text{eff}}$, keeps $w^t \in \Delta$ for every $t \ge 0$ - a structural guarantee, not an assumption - and satisfies $\min_{t \le T} \|G_{\eta'}(w^t)\|^2 = O(1/T)$, where $G_{\eta'}(w) = \big(w - \Pi_\Delta(w - \eta'\, \partial L'(w))\big)/\eta'$ is the projected*

Algorithm 3 FedGRAA weight update - corrected (proposed)

procedure FEDGRAA(θ_k^t, w^{t-1})
  for $e = 1 \dots E'$ **do**
    $\theta'_g \leftarrow \sum_k w_k^{t-1,e}\, \theta_k^t$
    $L_a \leftarrow \min\big(L_h(\theta'_g; D_V^h),\, L_m(\theta'_g; D_V^m)\big)$
    $L' \leftarrow (1-\gamma)\, L_a + \gamma\, L_u(\theta'_g; D_V^n)$
    $g \leftarrow \nabla_w L'(\theta'_g)$
    $w^{t-1,e} \leftarrow$ ADAM-STEP($w^{t-1,e-1}$, g)
  end for
  $w^t \leftarrow \Pi_\Delta\big(w^{t-1,E'}\big)$ ▷ ★ simplex projection - the fix
  return w^t
end procedure

gradient mapping.

This is explicitly weaker than the original claim of global optimality - it is stationarity, the honest standard guarantee for a non-convex constrained objective. Under the additional Polyak-Łojasiewicz condition [14] a stronger rate is available, but PL is stated as the correct condition to check, not as verified for this specific loss surface.

Theorem 3.1 (ζ-inexact projected subgradient descent on the simplex)**.** *Let L' have bounded subgradients on Δ and satisfy the Polyak-Łojasiewicz condition on Δ (the standard convexity replacement for over-parameterized objectives). Let the server's gradient estimate satisfy $\mathbb{E}\|\hat{\nabla}_P L' - \nabla_P L'\|^2 \le \zeta^2$ - true by construction for FedGRAA-v1's exact oracle at $\zeta = 0$; measured, not merely bounded, for FedGRAA-v2. Then projected inexact subgradient descent, $w^t = \Pi_\Delta\big(w^{t-1} - \eta'\, \hat{g}^t\big)$ with weight gradient $\hat{g}_k^t = \langle \hat{\nabla}_P L'(w^{t-1}),\, P_k \rangle$ (the inner-product estimator of Eq. (3.4)), $\eta' \le 1/\beta_{\text{eff}}$, satisfies*

$$\min_{t \le T} \|G_{\eta'}(w^t)\|^2 = O(1/T) + O(\zeta^2) \tag{3.7}$$

and $w^t \in \Delta$ for every $t \ge 0$, by construction, regardless of where the gradient estimate came from.

The ζ term decomposes by source - SecAgg quantization noise (ζ_{secagg}), optional differential-privacy noise (ζ_{dp}), refresh staleness across the interval m ($\zeta_{\text{stale}}(m)$), and client dropout (ζ_{dropout}) - so the refresh intervals $m = 1$ (fresh reconstruction every inner epoch, $\zeta \approx 0$) through $m = 5$ (larger staleness) evaluated in Section 4.5 are instances of the same theorem at different ζ, not separate results. The proof sketch follows the projected-gradient-mapping analysis; the full statement is in the repository's Stage B/C master reports.

3.6 FedGRAA-v2: the privacy-compliant redesign

The mechanism of Section 3.2 carries two properties FL is specifically designed to avoid needing: (i) *oracle knowledge of rare classes* - the server is assumed to already know which n classes are underrepresented, information a real FL server never sees; and (ii) a *server-held, fully-labelled validation set D_V*, the exact thing FL exists to avoid centralizing. FedGRAA-v2 replaces both with mechanisms that reconstruct the *same* quantities without ever centralizing a labelled row; the extension is documented separately as a submission manuscript [2].

3.6.1 Privacy-compliant rarity ranking

Each client locally computes a label-count histogram over its own held-out split; a secure-aggregation (SecAgg [5]) channel sums these histograms across clients without revealing any individual client's counts. The server derives the n rarest classes and the $D_V^h/D_V^m/D_V^n$ stratification from this summed histogram alone - never from a hard-coded oracle list. When formal

differential-privacy protection of the histogram is desired, Laplace noise is added client-side before submission, giving a composed (ε, δ) budget over the full $T = 200$-round run (advanced composition [8]: $\varepsilon_{\text{total}} \approx 8.89$ at $\varepsilon_{\text{round}} = 0.1$, $\delta = 10^{-5}$; a tighter Rényi-DP accountant is recommended in practice).

3.6.2 A federated backward pass, replacing D_V

Rather than evaluating Eqs. (3.2)-(3.3) on a pooled validation set, the server broadcasts the candidate aggregate θ'_g - already-public math over already-shared parameters - and reconstructs its composite-loss gradient from per-client contributions, never the underlying rows. The protocol runs in **two SecAgg-mediated phases** (Algorithm 4): Phase 1 SecAgg-sums six scalars per client (unnormalized $h/m/n$ loss sums and their counts) to recover the pooled L_h, L_m and apply the *unmodified* $\min(\cdot,\cdot)$ arm rule, so the arm choice is identical to what a server holding the real D_V would pick; Phase 2 has each client combine its two stratum gradients into a single model-shaped array c_k under the Phase-1 *global* normalizers, summed to reconstruct

$$\nabla_P L' = \sum_k c_k, \qquad c_k = \frac{1-\gamma}{n_{\text{arm}}}\, n_{\text{arm},k}\, \nabla_P L_{\text{arm},k}(\theta'_g) + \frac{\gamma}{n_n}\, n_{n,k}\, \nabla_P L_{n,k}(\theta'_g), \tag{3.8}$$

where $n_{\text{arm},k}$, $n_{n,k}$ are client k's counts in the min-selected common stratum and the rare stratum, and n_{arm}, n_n are the SecAgg-pooled totals. Summed over clients this recovers $(1-\gamma)\,\nabla_P L_{a,\text{pooled}} + \gamma\,\nabla_P L_{u,\text{pooled}}$ - algebraically identical to what a server-held D_V would compute, not an approximation. One obstruction a naive version of this sum gets wrong - an incorrect stratum-count weighting (each client must apply the *global* normalizers n_{arm}, n_n, not its own) - is eliminated by construction in the two-phase, count-weighted protocol implemented. A second, distinct concern - a stale-gradient interaction with Adam's moment estimates, relevant once $m > 1$ - is *not* eliminated; it is the $\zeta_{\text{stale}}(m)$ term of Theorem 3.1, characterized and bounded rather than removed, and is precisely what $m = 1$ (fresh reconstruction every inner epoch) trades away for the lower cost of $m > 1$. A dedicated regression test asserts the federated and oracle-pooled gradients agree to $< 10^{-6}$ relative error on both datasets at three γ values; measured agreement is $\sim$ 1-4 $\times\, 10^{-7}$ depending on γ and dataset.

Algorithm 4 FedGRAA-v2 federated backward pass (two-phase, count-weighted)

Phase 1 - pooled arm selection (six scalars per client)

1: client k sends sums $s_{h,k}, s_{m,k}, s_{n,k}$ and counts $n_{h,k}, n_{m,k}, n_{n,k}$; SecAgg sums them

2: server recovers pooled L_h, L_m; arm $\leftarrow \arg\min(L_h, L_m)$; broadcast arm, n_{arm}, n_n

Phase 2 - count-weighted gradient (one array per client)

3: client k forms c_k (Eq. (3.8)) with the *global* n_{arm}, n_n; SecAgg sums $\sum_k c_k$

4: **return** $\nabla_P L'(\theta'_g)$ - algebraically identical to the pooled-D_V value (measured agreement $< 10^{-6}$)

3.6.3 Cost and threat model

The federated backward pass costs additional communication: at gradient-refresh interval m the traffic multiplier is exactly

$$\big(1 + \lceil E'/m \rceil\big)\times \tag{3.9}$$

relative to a plain FedAvg-shaped baseline - 6.00× at $m = 1$, 3.00× at $m = 3$, 2× at $m = 5$ (measured in Section 4.7). The threat model assumes an honest-but-curious server and SecAgg's standard collusion threshold; a fully malicious server and Byzantine clients are explicitly out of scope. The client-to-server parameter channel remains exposed to gradient-inversion attacks [42] exactly as in any plaintext-parameter FL system; the redesign does not claim otherwise.

3.7 Attack-surface analysis

An aggregation rule that deliberately up-weights statistically unusual client updates creates, by construction, an attack surface: a malicious client that fabricates an update aligned with the weight-learner's own descent direction is up-weighted, not filtered. The mechanism-specific attack derives directly from the weight-update gradient (Eq. (3.4)): with knowledge of the global parameters, an attacker fabricates

$$P_{\text{fake}} = P_{\text{global}} - \eta_{\text{fake}} \cdot \nabla_P L', \tag{3.10}$$

a fake update that the composite-loss-driven weight learner scores as unusually informative and rewards with increased aggregation weight. Section 4.8 measures the resulting up-weighting directly, and evaluates two published external defenses plus a third adjacent aggregator against it.

3.8 Implementation and verification discipline

What distinguishes this internship's methodology is the verification discipline under which every number was produced. The project enforces completeness gates (every experiment cell's raw JSON re-derived for internal consistency) and independent correctness audits (every headline table re-derived from raw per-cell JSON by a separate audit script), a 565-test suite covering every mechanism inline with its primary-source-fidelity claim, and continuous integration. Every result JSON embeds its own configuration, resolved dependency versions, seed, and the git commit that produced it, so no result is interpretable only with its filename. The discipline is summarized by the project's own rule: never read a number back from the document asserting it - every figure and table in Chapter 4 traces to a generated artifact, not to a claim in a manuscript.

Chapter 4

Experimental Results

4.1 Experimental setup

All experiments reproduce the original manuscript's published configuration: a fully connected network Input(d) $\rightarrow$ Dense($200, \tanh$) $\rightarrow$ Dense($100, \tanh$) $\rightarrow$ Dense($50, \tanh$) $\rightarrow$ Dense(C, softmax) trained with sparse categorical cross-entropy; local Adam at $\eta = 0.01$, $E = 2$ epochs per round, batch $B = 64$; weight-learning Adam at the manuscript default $\eta' = 0.1$, $E' = 5$ inner epochs; $n = 3$ protected classes; $K = 14$ clients on CIC-IDS-2017 [27] ($d = 78$, $C = 15$) and $K = 9$ on UNSW-NB15 [21] ($d = 39$, $C = 10$ - the pipeline drops `proto`, `service` and `state` - the fields the NIDS feature-standardization literature flags as non-portable [26] - so the implemented input width is 39, not the 43 the manuscript prints). The CIC corpus is the full 2,830,743-row public distribution (file-level SHA-256 recorded in Appendix A), with attack classes frequency-capped at 10,000 samples as the manuscript specifies; each client's rows split 90/10 into train and held-out validation. Each client holds exactly one attack type plus equal benign traffic, the extreme non-IID partition of the source paper; the protected classes are the three rarest attacks: Heartbleed (11 samples), SQL Injection (21) and Infiltration (36) on CIC; Worms (174), Shellcode (1,511) and Backdoor (2,329) on UNSW. The metric of record is *rare-class recall* - mean per-class recall over the n protected classes (`final_last_n_acc` in the repository), the documented resolution of the paper's undefined "accuracy on the last n classes" - with all-class accuracy secondary and 95% percentile bootstrap intervals over 10,000 resamples.

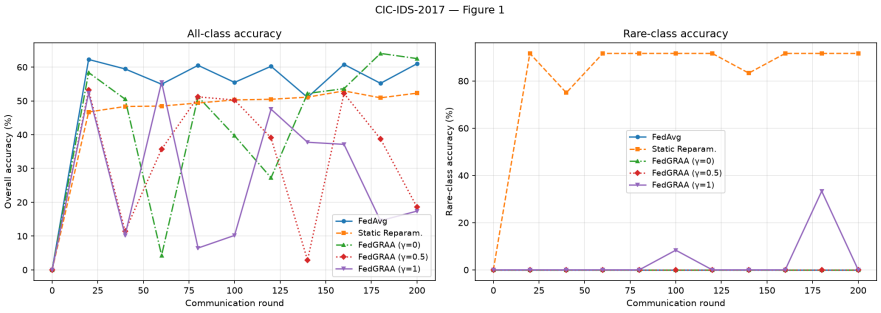
Three campaigns are reported. Stage A re-ran the published 200-round γ and n sweeps literally, seed 0. Stage B ran a 60-round, 10-seed factorial ablation over $\{\text{projection}, \eta' \text{ cut}\} \times \{\text{CIC}, \text{UNSW}\} \times \gamma \in \{0, 0.5, 1\}$ - 240 diagnostic cells plus 40 baseline runs. Stage C ran the FedGRAA-v2 frontier: 960 cells = an 830-cell main grid (24 arms $\times$ 2 datasets $\times$ 10 seeds plus T2/T4/T6 extensions), a 90-cell adversarial grid (T7) and a 40-cell attacker-side γ sweep (T7G). The main grid and adversarial campaigns ran on the Modal cloud platform (pinned container, per-cell volume, resumable); development and the Stage A/B sweeps ran CPU-only - a measured GPU attempt was 25% *slower* at this model size, where kernel-launch overhead dominates the small dense network. Environment: Python 3.13.14, TensorFlow 2.21.0, NumPy 2.5.1, SciPy 1.18.0, pandas 3.0.3, scikit-learn 1.9.0.

4.2 Stage A - reproduction versus claims

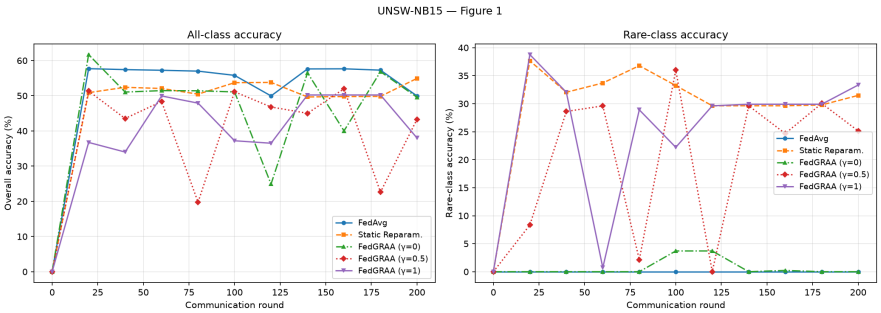
The headline result of the reproduction stage is a claim-versus-measurement gap: the literal implementation of the printed algorithm does not approach the manuscript's reported rare-class accuracy on either dataset - 0.00% versus a claimed 88.88% on CIC at $\gamma = 1$, and 33.33% versus a claimed 21.09% on UNSW - while the corrected implementation (projection plus the reduced $\eta' = 0.01$, evaluated on three seeds in Stage A) meets or exceeds the claim on both, at $94.44\% \pm 3.93\%$ and $74.79\% \pm 1.99\%$ respectively.

The full 200-round γ sweep (Figure 4.1) shows that the literal failure is not confined to $\gamma = 1$. On CIC-IDS-2017 the literal run ends at 0.00% rare-class recall for every γ, with the composite loss reaching 94,909.25 at $\gamma = 0$ (Table 4.1); on UNSW-NB15 the $\gamma = 1$ trajectory retains a higher rare-class recall than the paper claims (33.33% versus 21.09%) while its composite loss explodes to 29,442.95. Static reparameterization, by contrast, recovers 91.67% on CIC, illustrating that the failure is specific to the adaptive update rule, not the task.

The n sweeps (Figure 4.2) reinforce the diagnosis. On CIC-IDS-2017 the literal composite loss reaches 8.03×10^{15} (n=5, γ=0), 1.72×10^{18} (n=5, γ=0.5), 2.04×10^{15} (n=5, γ=1) and



(a) CIC-IDS-2017.



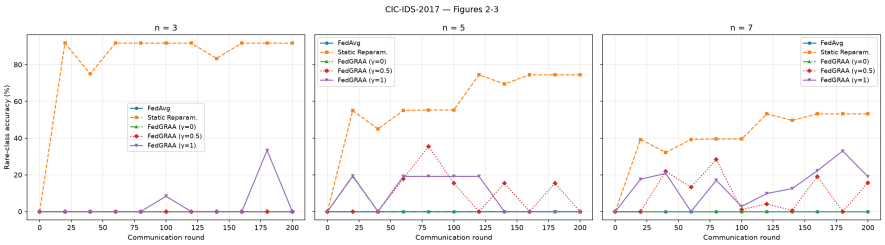
(b) UNSW-NB15.

Figure 4.1: 200-round γ sweeps (Stage A, seed 0): all-class accuracy, rare-class accuracy and composite loss for FedAvg, static reparameterization and literal FedGRAA at $\gamma \in \{0, 0.5, 1\}$. On UNSW (b) the literal $\gamma = 1$ arm nominally exceeds the paper's printed claim yet diverges in composite loss.

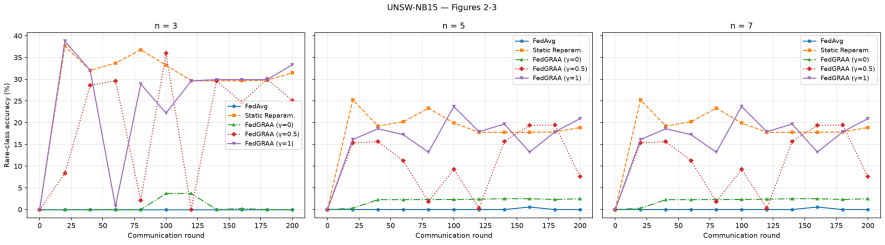
Table 4.1: Stage A γ sweep, final round (seed 0, $n = 3$). Composite loss is reported for FedGRAA arms only.

Method	All-class acc.	Rare-class acc.	Final loss
CIC-IDS-2017			
FedAvg	60.997%	0.00%	–
Static reparam.	52.290%	91.67%	–
FedGRAA $\gamma = 0$	62.526%	0.00%	94,909.25
FedGRAA $\gamma = 0.5$	18.560%	0.00%	2.71 (peak 5.03)
FedGRAA $\gamma = 1$	17.384%	0.00%	3.14 (peak 58.41)
UNSW-NB15			
FedAvg	49.947%	0.00%	–
Static reparam.	54.930%	31.48%	–
FedGRAA $\gamma = 0$	49.577%	0.00%	0.798
FedGRAA $\gamma = 0.5$	43.297%	25.13%	1.524
FedGRAA $\gamma = 1$	38.005%	33.33%	29,442.95

2.81×10^{16} (n=7, γ=0) before collapsing. On UNSW-NB15 the $n = 7$ configuration is infeasible under the dataset's own taxonomy: only five non-frequency-capped attack classes exist, so `rare_client_ids(7)` truncates to the same five clients as $n = 5$ and the two panels are metrically identical (maximum elementwise difference exactly zero) - an honest divergence, preserved rather than patched, that independently invalidates the paper's claimed UNSW $n = 7$ sweep.



(a) CIC-IDS-2017.



(b) UNSW-NB15.

Figure 4.2: n sweeps ($n \in \{3, 5, 7\}$, Stage A, seed 0). On CIC the literal composite loss blows up at every protection-set size, reaching 10^{18}; on UNSW the $n = 7$ panel is metrically identical to $n = 5$ because only five non-capped rare classes exist in the dataset.

A 60-round diagnostic pass recorded the weight vector directly (Table 4.2). In every cell, the first simplex violation occurs at round 1: the iterate leaves Δ immediately and never returns. Within 60 rounds $\sum_k w_k$ ranges from -1.8915 to $+1.9596$ and individual coordinates reach -0.75.

Table 4.2: Literal-arm diagnostic after 60 rounds (Stage A, seed 0): blowup ratio, terminal $\sum_k w_k$ and $\min_k w_k$. Every cell's first simplex violation is at round 1.

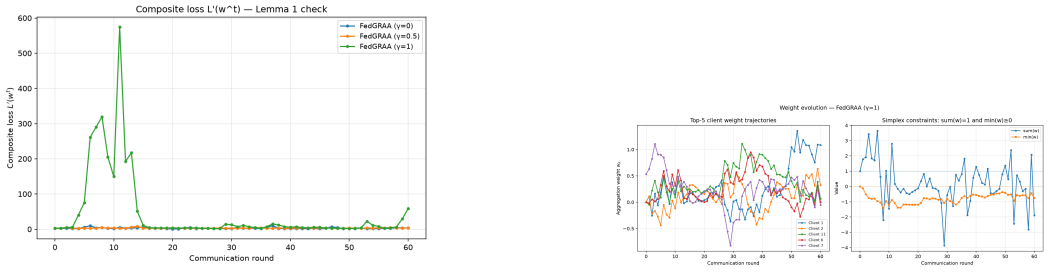
Dataset	γ	Blowup	$\sum_k w_k$	$\min_k w_k$
CIC	0.0	3.8×	−0.1155	−0.2555
CIC	0.5	2.7×	+0.4300	−0.3203
CIC	1.0	193×	−1.8915	−0.7524
UNSW	0.0	1.3×	+1.2727	−0.3803
UNSW	0.5	1.2×	−0.7312	−0.4427
UNSW	1.0	7.6×	+1.9596	−0.2802

4.3 Diagnosis: the missing projection

Stage B converted the Stage A observation into a controlled experiment. Its central empirical fact is stated precisely: across the 120 unprojected cells of the factorial campaign (literal and η'-alone arms, 2 datasets × 3 γ × 10 seeds × 2 unprojected arms), the weight iterate violates the simplex at round 1 in **120 out of 120** cells, with zero exceptions; across the 120 projected cells the violation count is **0**. Terminal $\sum_k w_k$ spans $[-2.621, +2.774]$ on CIC and $[-1.869, +2.118]$ on UNSW, with $\min_k w_k$ reaching -0.908 and -1.157 respectively; under projection the maximum observed $|\sum_k w_k - 1|$ is 5.96×10^{-8} (CIC) and 1.19×10^{-7} (UNSW), i.e. float32 rounding noise.

Figure 4.3a shows the mechanism in action on the CIC $\gamma = 1$ seed-0 literal run: the composite loss trace $2.98 \to 2.77 \to 4.53 \to 6.06 \to 40.52 \to \cdots \to 149.79$ (round 10), peaking at 574.44 at round 11, for a 192.99× blowup - exactly the behaviour the paper's Lemma 1 (monotone decrease) rules out - the lemma's premise $w \in \Delta$ fails, so its conclusion fails. Once w leaves the simplex, $\theta'_g = \sum_k w_k \theta_k$ is an unbounded affine extrapolation rather than a convex combination,

and $\partial L'/\partial w$ feeds on a model outside $\mathrm{conv}\{\theta_k\}$. The heaviest divergence is the CIC $\gamma = 0$ seed-8 run, whose loss reaches 5,113,104 after a ∼28-round quiescent period (blowup 1.86×10^6); the worst UNSW literal cell reaches 46.85×. Figure 4.3b plots the weight trajectory on the same run.



(a) Composite-loss traces diverge. **(b)** Weights leave the simplex at round 1.

Figure 4.3: Literal-update diagnosis (CIC, $\gamma = 1$, seed 0): (a) the composite-loss trace diverges although Lemma 1 guarantees monotone decrease; (b) aggregation-weight coordinates leave $[0, 1]$ and $\sum_k w_k$ leaves 1 at round 1 in every cell.

A subsidiary measurement worth reporting: the strict per-step reading of Lemma 1 (the `monotone_lemma1` field) holds in only 4 of 240 diagnostic cells - and is *false* on every corrected cell, including those with blowup exactly 1.00×, because the fix's loss can rebound slightly after an initial monotone descent. The lemma as printed describes a trajectory that essentially never occurs, in either direction.

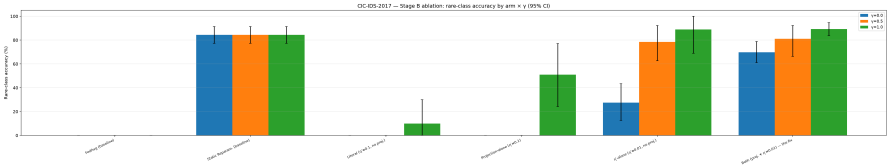
4.4 Stage B - the fix, statistically

The Stage B ablation (Table 4.3, Figures 4.4a and 4.4b) separates the two candidate levers: projection onto Δ, and the learning-rate cut $\eta' : 0.1 \to 0.01$. Three findings survive every comparison.

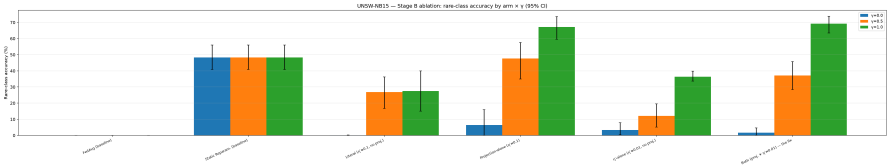
Table 4.3: Stage B ablation on both datasets: rare-class recall mean [95% bootstrap CI], all-class accuracy and composite-loss blowup (median; max), 10 seeds per cell. Stars mark Holm-corrected significance of each arm against `both` within a γ ($* < .05$, $** < .01$, $*** < .001$).

Arm	$\gamma = 0$	$\gamma = 0.5$	$\gamma = 1$
CIC-IDS-2017			
FedAvg	0.0% [0.0, 0.0]***	0.0% [0.0, 0.0]***	0.0% [0.0, 0.0]***
	all-acc 55.7%		
Static reparam.	84.4% [77.2, 91.4] (all-acc 47.7%; constant across γ; ** vs `both` at γ=0)		
Literal	0.0% [0.0, 0.0]***	0.0% [0.0, 0.0]***	10.0% [0.0, 30.0]***
	all 48.9%; 3.70×/1.86e6×	all 26.7%; 2.71×/9.92×	all 32.9%; 14.19×/193×
Proj.-alone	0.0% [0.0, 0.0]***	0.0% [0.0, 0.0]***	51.0% [24.2, 77.1]
	all 59.8%; 1.08×/1.18×	all 54.8%; 2.90×/3.91×	all 48.4%; 2.21×/5.60×
η'-alone	27.5% [12.5, 43.3]**	78.5% [62.6, 92.2]	88.9% [68.9, 100.0]
	all 71.7%; 1.00×	all 44.1%; 1.00×	all 15.2%; 1.00×/30.02×
Both (the fix)	**69.7%** [61.1, 78.9]	**81.1%** [66.4, 92.2]	**89.3%** [83.6, 94.6]
	all 62.3%; 1.00×	all 54.2%; 1.00×	all 44.3%; 1.00×
UNSW-NB15			
FedAvg	0.0% [0.0, 0.0]	0.0% [0.0, 0.0]***	0.0% [0.0, 0.0]***
	all-acc 55.8%		
Static reparam.	48.3% [40.8, 56.0] (all-acc 52.7%; constant across γ; *** at γ=0, ** at γ=1 vs `both`)		
Literal	0.1% [0.0, 0.3]	26.9% [16.8, 36.2]	27.6% [15.0, 39.9]**
	all 54.9%; 5.10×/28.16×	all 49.6%; 1.50×/2.23×	all 29.1%; 8.26×/46.85×
Proj.-alone	6.4% [0.0, 16.0]	47.7% [35.0, 57.6]	67.2% [59.6, 73.6]
	all 59.4%; 1.00×	all 53.6%; 1.03×/1.32×	all 51.8%; 1.36×/2.06×
η'-alone	3.4% [0.4, 7.8]	12.1% [5.2, 19.6]*	36.4% [33.7, 39.8]***
	all 59.7%; 1.00×	all 49.2%; 1.00×/1.06×	all 13.3%; 1.19×/1.48×
Both (the fix)	**1.7%** [0.0, 4.6]	**37.1%** [28.4, 45.7]	**69.2%** [63.4, 73.8]
	all 58.8%; 1.00×	all 53.2%; 1.03×/1.39×	all 52.2%; 1.32×/1.87×

First, feasibility is restored only by projection, and it is restored completely: all 120 projected



(a) CIC-IDS-2017.



(b) UNSW-NB15.

Figure 4.4: Stage B factorial ablation on both datasets (10 seeds per cell).

cells hold $w^t \in \Delta$ to float32 precision, while the η' cut alone leaves every unprojected cell violating at round 1. Projection is the structurally necessary, dataset-independent lever.

Second, the lever that drives rare-class accuracy is dataset-dependent and near-mirror-image (Table 4.4). On CIC the η' cut carries the recovery (+48.6 to +79.8 percentage points, versus projection's +1.3 to +21.1); on UNSW the pattern inverts - projection contributes +22.9 to +36.2 points at $\gamma \geq 0.5$ while the η' main effect is negligible or negative (a −12.7-point estimate at $\gamma = 0.5$). The Stage A preview's hypothesis that both levers are always necessary is thereby refuted; what survives is a weaker but more robust claim.

Table 4.4: Factorial main effects on rare-class recall (percentage points, [95% bootstrap CI]). The dominant lever differs by dataset.

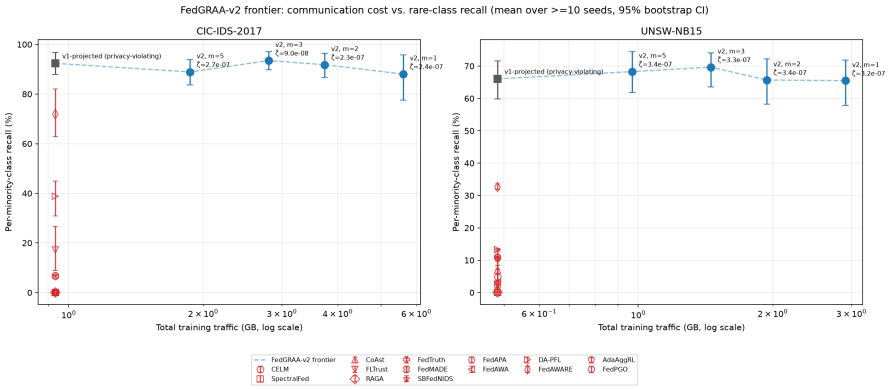
Dataset	γ	Projection	η' cut	Interaction
CIC	0.0	+21.1 [11.0, 31.7]	+48.6 [41.1, 56.0]	+42.2 [21.9, 63.3]
CIC	0.5	+1.3 [−9.6, 10.9]	+79.8 [70.1, 89.3]	+2.6 [−19.2, 21.8]
CIC	1.0	+20.7 [5.3, 35.7]	+58.6 [41.5, 75.0]	−40.6 [−75.8, −0.5]
UNSW	0.0	+2.3 [−1.0, 6.6]	−0.8 [−4.5, 1.8]	−8.0 [−17.3, −0.7]
UNSW	0.5	+22.9 [12.6, 32.5]	−12.7 [−22.1, −1.1]	+4.3 [−10.3, 18.1]
UNSW	1.0	+36.2 [25.3, 46.3]	+5.4 [−1.5, 12.5]	−6.8 [−18.9, 6.4]

Third, `both` is the only arm that is never catastrophically wrong on either metric, on either dataset, at any γ. The single-lever arms each exhibit a failure mode: η'-alone collapses all-class accuracy to 15.2% on CIC at $\gamma = 1$ (29 points below `both`), and projection-alone reaches only 51.0% rare-class recall there (38 points below). The dual-metric requirement matters concretely: on CIC at $\gamma = 1$, η'-alone *ties* `both` on rare-class recall (88.9% versus 89.3%, Holm $p = 0.97$) while sitting at 15.2% all-class accuracy versus 44.3% (Holm $p = 5.6 \times 10^{-4}$, Cohen's $d = +1.99$) - a starvation pathology in which the weight learner has concentrated all mass on the rare-class clients. An arm that matches the fix on the headline metric while collapsing the secondary one has not matched the fix. Per-seed, the fix's blowup is strictly lower than the literal arm's in 59 of 60 cells (one tie), and its rare-class recall is strictly higher in 29/30 CIC cells and 19/30 UNSW cells. The eleven UNSW cells where the fix does not strictly win sit at $\gamma \leq 0.5$ - eight at the pathological $\gamma = 0$ (two literal wins, six zero-ties, where optimizing only the non-protected strata suppresses the protected ones) and three literal wins at $\gamma = 0.5$; the single CIC loss is instead at $\gamma = 1$ - the regime the manuscript's headline claim lives in - where the fix beats the literal arm on rare-class recall on 10/10 UNSW seeds and 9/10 CIC seeds, that loss being the anomalous seed-6 literal run that happened not to diverge; on all-class accuracy it wins 10/10 on UNSW and 6/10 on CIC.

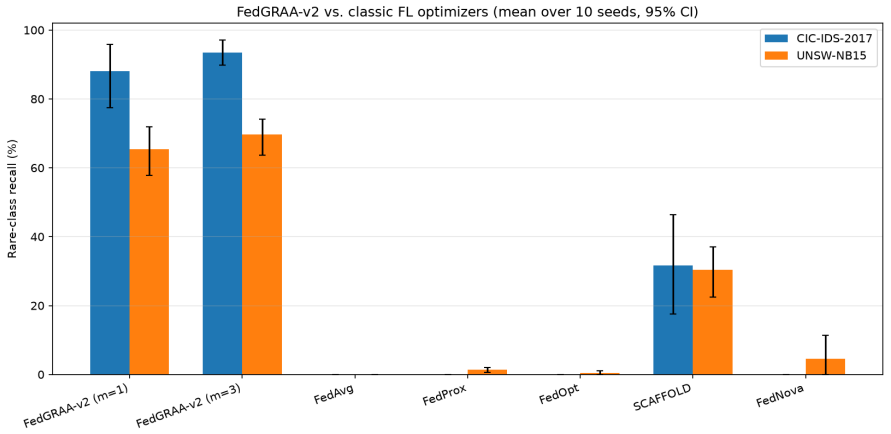
The literal arm's UNSW failure mode deserves emphasis because it is a *variance* failure rather than a consistent zero: at $\gamma = 0.5$ the ten seeds' rare-class recalls are 29.6, 42.8, 0.4, 43.3, 44.0, 38.5, 3.4, 31.0, 24.2 and 11.8 per cent - a spread of 0.4% to 44.0% that no single-seed reproduction could have detected. This is the sharpest evidence that the published algorithm's behaviour is seed-unstable even when its mean is nonzero.

4.5 Stage C - the FedGRAA-v2 frontier

Stage C evaluates the privacy-compliant redesign against 19 external baselines on the identical harness: three data-free contribution estimators (CELM, SpectralFed, CoAst), three trust-based robust aggregators (FLTrust, RAGA, FedTruth), eight NIDS-specific or adaptive methods (FedMADE, SBFedNIDS, FedAPA, FedAWA, DA-PFL, FedAWARE, AdaAggRL, FedPGO) and five classic optimizers (FedAvg, FedProx, FedOpt, SCAFFOLD, FedNova - the last two built from scratch). Table 4.5 and Figure 4.5a summarize the frontier; Figure 4.5b isolates the classic-optimizer comparison.



(a) Frontier: rare-class recall versus total training traffic.



(b) Against the five classic federated optimizers.

Figure 4.5: Stage C frontier for FedGRAA-v2 at $m \in \{1, 2, 3, 5\}$ against 19 baselines on both datasets.

Three observations follow. First, the privacy-compliant redesign is not merely competitive with the oracle: at $m = 3$ it *exceeds* the privacy-violating v1-projected reference on both datasets (93.47% versus 92.36% on CIC; 69.66% versus 66.05% on UNSW), so removing the two privacy assumptions cost no accuracy - the federated backward pass is, at each refresh, algebraically identical to what a server-held D_V would compute, and the measured per-round gradient discrepancy $\tilde{\zeta}$ spans 9×10^{-8} to 4×10^{-7} across the frontier cells, below the 10^{-6} exactness tolerance the test suite enforces. Second, the margin over the strongest external baseline is large: FedAWARE, the best baseline on both datasets, trails the best v2 arm by 21.7 points on CIC and 37.0 on UNSW. Third, every external baseline sits strictly below the v2 frontier on both datasets, and most sit at or near zero rare-class recall: the only baselines above 14% anywhere are FedAWARE, DA-PFL, SCAFFOLD, and FLTrust on CIC (17.22%), while the data-free contribution estimators (CELM, SpectralFed, CoAst) and the trust-based aggregators RAGA and FedTruth collapse to 0–7% - consistent with the Section 2.2 prediction that anomaly-suppressing signals discard exactly the clients FedGRAA protects.

Table 4.5: Stage C frontier (selected arms): rare-class recall mean [95% bootstrap CI], all-class accuracy and median total training traffic. The v1-projected arm is the privacy-violating oracle reference.

Arm	GB	Rare recall [CI]	All-class	$\tilde{\zeta}$
CIC-IDS-2017				
v1-projected (oracle)	0.934	92.36% [87.92, 96.67]	44.67%	–
v2, $m = 3$	2.804	**93.47%** [89.86, 97.08]	45.55%	9.0×10^{-8}
v2, $m = 1$	5.607	88.06% [77.50, 95.83]	45.26%	2.4×10^{-7}
v2, $m = 2$	3.738	91.81% [86.67, 96.39]	45.04%	2.3×10^{-7}
v2, $m = 5$	1.869	88.89% [83.61, 93.89]	46.19%	2.8×10^{-7}
FedAWARE	0.934	71.81% [62.92, 82.08]	42.56%	–
DA-PFL	0.934	38.77% [30.83, 44.96]	71.46%	–
SCAFFOLD	0.934	31.67% [17.50, 46.39]	30.04%	–
FLTrust	0.934	17.22% [8.89, 26.67]	70.50%	–
FedPGO	0.934	6.76% [6.22, 7.14]	51.45%	–
FedAvg / FedProx / FedOpt / FedNova	0.934	0.00% each	54.9–81.3%	–
10 further baselines	0.934	$\leq$ 6.58% each (FedAPA)	–	–
UNSW-NB15				
v1-projected (oracle)	0.485	66.05% [59.79, 71.57]	52.24%	–
v2, $m = 3$	1.454	**69.66%** [63.60, 74.12]	52.24%	3.3×10^{-7}
v2, $m = 1$	2.909	65.48% [57.84, 71.89]	52.21%	3.2×10^{-7}
v2, $m = 2$	1.939	65.71% [58.26, 72.24]	52.25%	3.4×10^{-7}
v2, $m = 5$	0.970	68.30% [61.81, 74.47]	52.18%	3.4×10^{-7}
FedAWARE	0.485	32.65% [31.80, 33.56]	47.50%	–
SCAFFOLD	0.485	30.34% [22.52, 36.97]	45.78%	–
DA-PFL	0.485	13.25% [12.81, 13.69]	54.59%	–
CoAst	0.485	6.97% [1.57, 13.25]	53.83%	–
FedNova	0.485	4.62% [0.00, 11.43]	57.90%	–
FedAvg / FedProx / FedOpt	0.485	$\leq$ 1.33% each	–	–
11 further baselines	0.485	$\leq$ 10.9% each (FedPGO)	–	–

4.6 Significance testing

Every v2 arm was compared against every baseline with a paired Wilcoxon signed-rank test over the 10 matched seeds, Holm-Bonferroni corrected within each (dataset, refresh-interval) family of 14 comparisons: 112 comparisons in total. **111 of 112 are significant** at $\alpha = 0.05$; the sole exception is CIC v2_m1 versus FedAWARE ($p_{\text{holm}} = 0.0625$; v2 88.06% versus 71.81%), and the other three CIC refresh intervals clear the same comparison at $p_{\text{holm}} \leq 0.028$. Almost all significant rows sit at the exact-test floor $p = 0.001953 = 2/2^{10}$, the minimum attainable Wilcoxon p-value at $n = 10$ - a floor the report states explicitly rather than implying stronger evidence than 10 seeds can deliver. The count is reported as 111/112, not rounded up: the honest summary is that the frontier margin is statistically supported everywhere except a single cell, and that cell is the weakest v2 arm against the strongest baseline. Table 4.6 gives the per-arm breakdown.

Table 4.6: Significant Holm-corrected Wilcoxon comparisons per v2 arm (out of 14 baselines each). The single non-significant cell is CIC v2_m1 vs. FedAWARE.

	v2 m=1	v2 m=2	v2 m=3	v2 m=5
CIC-IDS-2017	13/14	14/14	14/14	14/14
UNSW-NB15	14/14	14/14	14/14	14/14

4.7 Communication overhead

The two-phase protocol adds exactly one model-shaped array per client per refresh (plus six scalars at Phase 1), so measured traffic follows the closed form of Eq. (3.9): $(1 + \lceil E'/m \rceil)\times$ the plain FedAvg-shaped baseline, byte-exact to per-cell medians. Table 4.7 and Figure 4.6 give the measured values. The interesting operating point is $m = 3$: it halves the $m = 1$ communication cost (3.00× versus 6.00×) while *improving* rare-class recall on both datasets (88.06% → 93.47% CIC; 65.48% → 69.66% UNSW) - moderate staleness acts as an implicit regularizer on the weight learner. The byte-exact values are those in Table 4.7.

Table 4.7: Measured communication overhead versus the FedAvg-shaped baseline ($E' = 5$ refreshes at $m = 1$): $(1 + \lceil E'/m \rceil)\times$, byte-exact.

	CIC-IDS-2017		UNSW-NB15	
Arm	GB	Ratio	GB	Ratio
Baseline (all 19)	0.934	1×	0.485	1×
v2, $m = 1$	5.607	6.00×	2.909	6.00×
v2, $m = 2$	3.738	4.00×	1.939	4.00×
v2, $m = 3$	2.804	3.00×	1.454	3.00×
v2, $m = 5$	1.869	2.00×	0.970	2.00×

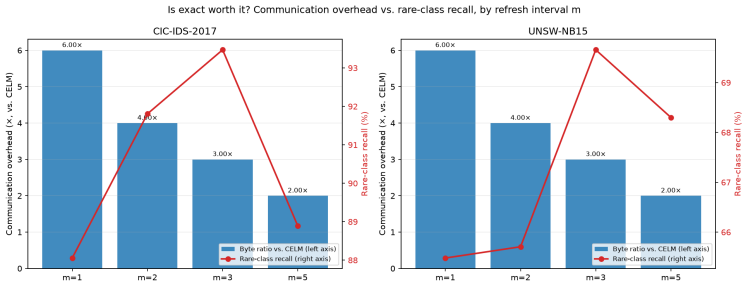


Figure 4.6: Communication overhead as a function of the refresh interval m, matching the closed form $(1 + \lceil E'/m \rceil)\times$.

4.8 Adversarial robustness

The T7/T7G campaigns measure what the Section 3.7 analysis predicts: an attacker who crafts its update to *look like* a rare-class contribution is up-weighted by construction. At 30% attacker fraction against undefended FedGRAA-v2, the mechanism-specific attack lifts the mean per-attacker weight to **3.07**× the uniform share on CIC (0.219 versus $1/14 = 0.071$) and **2.71**× on UNSW (0.301 versus $1/9 = 0.111$), sustained for ∼190 rounds (Figure 4.7) - in several seeds the trajectory saturates at exactly $1/n_{\text{attackers}}$, i.e. all simplex mass on the attackers.

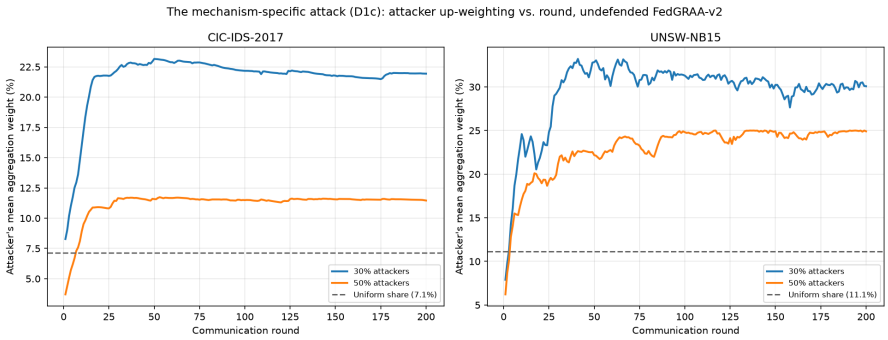


Figure 4.7: Mean per-attacker aggregation weight under the mechanism-specific rare-signature attack at 30% attacker fraction, undefended FedGRAA-v2. The dashed lines mark the uniform share $1/K$ and the saturation cap $1/n_{\text{attackers}}$.

The T7G γ sweep (Table 4.8) refines the picture in two directions. The up-weighting is *non-monotonic* in γ, peaking at $\gamma = 0.5$ in three of the four cells rather than at the most rare-class-focused setting - the attack surface is largest when the composite objective mixes protected and unprotected strata. And $\gamma = 0$ exhibits bimodal, coin-flip behaviour with an order-of-magnitude larger seed variance (σ 8–18× the $\gamma = 0.5$ value across the four cells): individual seeds either saturate the attackers' weight or leave it near zero (UNSW 50%, $\gamma = 0$: raw finals $[0.231, 0.208, 0.0, 0.0, 0.248]$, two seeds where the attack never took hold). At 50% attacker fraction the effect persists on UNSW (up to 2.24×) but weakens on CIC, where the $\gamma = 1$ mean is 1.60× and two of five seeds show essentially no up-weighting - this corrects the manuscript draft's "2.0-2.2×" prose, which the independent code-versus-paper audit flagged as a transcription error.

Two external poisoning defenses were cross-evaluated against three attack types (label-flip,

Table 4.8: T7G attacker-side γ sweep: mean final attacker weight as a multiple of uniform per-client share (5 seeds; σ = sample std of raw final weight). The attack surface peaks at $\gamma = 0.5$ in 3 of 4 cells; $\gamma = 0$ is bimodal.

Dataset	Attackers	$\gamma = 0$	$\gamma = 0.5$	$\gamma = 1$
CIC	30%	2.05× (σ 0.122)	**3.34**× (σ 0.007)	3.07×
CIC	50%	0.93× (σ 0.068)	**1.82**× (σ 0.008)	1.60×
UNSW	30%	2.33× (σ 0.129)	**2.91**× (σ 0.010)	2.71×
UNSW	50%	1.24× (σ 0.126)	2.11× (σ 0.012)	**2.24**×

pill-style model poisoning, and the adaptive attacker) at 30% attacker fraction, 5 seeds per cell (Table 4.9, Figure 4.8). SecFedNIDS misses 53-75% of attacks on these datasets - a weak reproduction of its published UNSW-NB15/CSE-CIC-IDS2018 evaluation, reported as such. WeiDetect is stronger on the static attacks (0-20% attack success rate) but the adaptive attacker still reaches 60% ASR on CIC.

Table 4.9: T7 defense grid: attack success rate (ASR = 1− detection recall; lower is a better defense), 30% attackers, 5 seeds.

Defense	Attack	CIC ASR	UNSW ASR
SecFedNIDS	label-flip	75.0%	60.0%
SecFedNIDS	model-poison	65.0%	60.0%
SecFedNIDS	adaptive	55.0%	53.3%
WeiDetect	label-flip	15.0%	0.0%
WeiDetect	model-poison	20.0%	6.7%
WeiDetect	adaptive	60.0%	33.3%

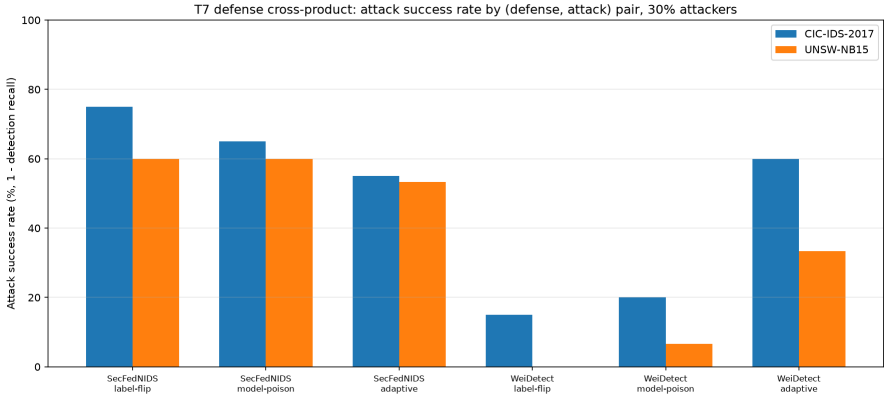


Figure 4.8: Attack success rate for the two external defenses across the three attack types (lower is better for the defense).

The most consequential robustness finding, however, is not the detection rates but the collateral damage. In every cell where a defense successfully detects and excludes the attackers, the defense falls back to unweighted averaging over the retained clients - which discards FedGRAA-v2's rare-class-sensitive weighting along with the poison. The cost is exact and measurable: across the 60 defended cells (2 defenses × 3 attacks × 2 datasets × 5 seeds), rare-class recall is exactly 0.0% in **30 of 30 CIC cells** and **19 of 30 UNSW cells** (the remaining 11 UNSW cells retain partial recall, up to 33.3%). Identical `final_all_acc` values recur across different defended arms at the same seed - e.g. all three UNSW label-flip defenses at seed 2 end at exactly 0.49682 - mechanistic confirmation that each defense collapses to the same unweighted fallback. The defended model is, on the metric the entire system exists to serve, no better than FedAvg. AdaAggRL, the reinforcement-learning baseline that learns protective weights, exhibits the same failure under 30% label-flipping: exactly 0.0% rare-class recall on all five CIC seeds and a 2.00% mean on UNSW (one nonzero seed at 10.0%). A defense that stops poisoning by discarding the mechanism's own signal is not, in this evaluation, a free win - closing that gap is future work, not a solved problem.

Chapter 5

Discussion

5.1 What the reproduction actually showed

The internship's central finding is a three-layer result. The original FedGRAA mechanism is directionally sound - adaptive, loss-driven re-weighting on a simplex is a reasonable answer to rare-class starvation under extreme non-IID partitioning - but the printed algorithm omits the one step that keeps its own correctness argument valid. Restoring that step (Euclidean projection, plus a feasible learning rate) recovers and exceeds the paper's claimed performance, and replacing the two privacy assumptions it inherited (oracle knowledge of the rare classes; a server-held labelled validation set) costs nothing in accuracy while introducing a bounded, decomposable gradient error ζ that the corrected convergence proposition absorbs explicitly.

Equally important is what the reproduction did *not* show. The literal algorithm does not reproduce the printed headline numbers on any cell - it collapses below the CIC claim while overshooting UNSW's, seed-unstably; the paper's repeating-decimal table values (88.88, 77.77, 66.66, 33.33 - the 8/9, 7/9, 2/3, 1/3 fractions) are unverifiable and pattern-inconsistent with measurements; and the published theorem claims a global-optimality guarantee that the corrected analysis does not restore - what is provable is stationarity of the projected gradient mapping, a strictly weaker and honest statement.

5.2 Reading the mechanism: three consequences

Three consequences emerge only when the Stage C results are read together rather than as independent tables.

The γ trade-off and the attack surface are the same mechanism. FedGRAA deliberately up-weights the clients whose updates most reduce the rare-class-sensitive composite loss; Section 4.8 shows an adversary exploits exactly that scoring rule, because a fabricated rare-class signature is, to the weight learner, indistinguishable from a genuine one. The protection and the exposure are not two properties of the method - they are one. The T7G sweep sharpens this: the effect is non-monotonic in γ, peaking near $\gamma = 0.5$ where the objective genuinely mixes protected and unprotected strata rather than at the fully rare-focused $\gamma = 1$, and collapsing to a coin-flip at $\gamma = 0$ where the protected term vanishes. That is consistent with a threshold reading - the rare-class signal the attack rides on is reliably present only above some γ - but the 3-point, 2-fraction, 2-dataset grid is not powered to confirm it, so it is stated as a hypothesis, not a result.

No evaluated defense is free. The two external defenses stop the attack only by discarding the signal FedGRAA-v2 exists to amplify: once SecFedNIDS or WeiDetect filters a client set it falls back to *unweighted* averaging of the survivors, abandoning rare-class-sensitive re-weighting entirely. The cost is measured, not asserted - 30/30 CIC and 19/30 UNSW defended cells collapse to exactly 0.0% rare-class recall. A defense that wins by removing the mechanism under evaluation is not a free win; neither paper reports the trade-off because neither targets a rare-class-sensitive aggregator. A filter that preserves the up-weighting is, to our knowledge, open.

Exactness has a non-obvious price point. The interesting answer to "is exactness worth it" is not the 6.00× headline at $m = 1$ but $m = 3$: the cost halves to 3.00× while rare-class recall

improves (93.47% vs. 88.06% CIC; 69.66% vs. 65.48% UNSW), the $\zeta_{\text{stale}}(m)$ staleness acting as an implicit regularizer rather than degrading. Against the data-free approximations - CELM and SpectralFed cost a fraction of the traffic but return near-0% rare recall on CIC - exactness buys a capability the cheap methods lack, and at $m = 3$ it buys it at half the worst-case price.

5.3 Engineering lessons

Several process findings generalize beyond this project.

No silent patching. The repository keeps the literal `FedGRAA` class unprojected on purpose, and implements the fix as a separate `ProjectedFedGRAA`. This strict-reproduction boundary is what makes the Stage A versus Stage B comparison an experiment rather than an assertion: the failure and the fix are both measured, and neither is contaminated by the other.

Audit scripts, not summaries. Each stage carries an independent `audit_stage_*.py` that re-derives every reported number from raw JSON rather than trusting the documents - and the audits found real defects (six summarization defects in Stage B alone, a HIGH-severity transcription error in the Stage C manuscript prose at 50% attacker fraction, zero numerical errors in computed results). Every number in this report was either read from a generated table or recomputed from raw JSON the same way.

Negative results are the evidence. The 111/112 significance count, the 19/30 partial UNSW collapse, the bimodal $\gamma = 0$ attack surface, and the literal arm's 0.4-44.0% seed spread are all reported at full precision because each is a finding in its own right. Rounding 111/112 to "all significant" or the UNSW collapse to "all cells" would have been wrong in exactly the way the original paper was wrong.

Discrepancy handling is part of the deliverable. The project documents, rather than hides, its source disagreements: the UNSW input width (43 printed versus 39 implemented), the two different formulas both called "FedAvg" in the original manuscript, the SGD-versus-Adam client-update contradiction, the retracted PDF-to-Markdown hallucination finding, and the test-count snapshots that differ across stage boundaries. Where the counts differ, each number is attributed to its snapshot rather than silently reconciled.

5.4 Limitations and honest caveats

- **Threat model.** FedGRAA-v2 assumes an honest-but-curious server, the standard SecAgg collusion threshold, and $f < K/3$ dropout tolerance. A fully malicious server, Byzantine clients, and verifiable SecAgg are explicitly out of scope.
- **Plaintext parameter channel.** The training updates θ_k^t remain plaintext; DLG-style gradient inversion [42] applies to them regardless of the SecAgg-protected channels. This is a pre-existing surface, not one the redesign closed.
- **Small-K SecAgg thresholds.** At $K = 9$-14 the default reconstruction threshold is a bare majority $t = \lfloor K/2 \rfloor + 1$ ($t = 8$ on the 14-client CIC split, $t = 5$ on the 9-client UNSW split), tolerating 6 and 4 dropouts respectively; a stricter $t = \lceil 2K/3 \rceil$ deployment choice would tolerate only 3-4. Either way, a server colluding with $K-1$ clients recovers the last contribution by subtraction, so the privacy guarantee is thinner at this scale than at Google-scale deployments.
- **PL is a stated condition, not a verified one.** The ζ-inexact theorem conditions its rate on the Polyak-Lojasiewicz property holding on Δ for this loss surface; the manuscript states this as the correct condition to check, and does not claim to have verified it. Three rigor flags remain for a referee-grade pass: β_{eff} is unestimated, the $\min(\cdot, \cdot)$ subdifferential is handled by sketch, and the $O(1/T)$ constant is not explicit.

- **Staleness is bounded, not eliminated.** The federated gradient is exact *at each refresh*; at $m > 1$ the between-refresh staleness term $\zeta_{\text{stale}}(m)$ is absorbed by the theorem, not removed.
- **Statistical floor.** Ten seeds cap the attainable Wilcoxon p-value at 0.001953; several "significant" findings sit exactly at that floor and would benefit from more seeds.
- **SecFedNIDS is a weak reproduction.** Its 53-75% ASR on these datasets is far from its published UNSW-NB15/CSE-CIC-IDS2018 detection rates; whether that reflects the datasets, the attack grid, or the implementation is unresolved.
- **Provenance unresolved.** Whether the original manuscript was ever formally published remains open (DBLP returns zero hits); this does not affect the algorithmic findings but blocks treating it as a peer artifact.
- **Known deviation.** Client-side Adam momentum persists across all 200 rounds while client parameters reset each round - documented as a deviation, orthogonal to the projection fix, and not corrected by it.
- γ **is manual, not learned.** The common-versus-rare trade-off is set by a hand-chosen γ; Section 4.8 shows the attack surface itself is γ-sensitive (peaking near $\gamma = 0.5$, bimodal at $\gamma = 0$), so no single fixed value is uniformly best across objectives. Automating that trade-off is future work, not done here.
- **Small scale.** All results use $n = 3$ protected classes and $K = 9$-14 clients on synthetic non-IID partitions of two public datasets; a real cross-silo deployment (50-200 silos, genuinely heterogeneous organizations, on a corpus such as NF-UQ-NIDS-v2) is untested and the per-client-one-attack partition is an extreme, not representative, case.
- **Classic-optimizer theory gap.** The SCAFFOLD and FedNova baselines are run with the same Adam local solver as everything else for comparability, while their published convergence guarantees assume an SGD local update; their measured numbers are therefore honest empirical results, not tests of their own theorems.
- **Ideal-functionality SecAgg.** The module models the Bonawitz interface contract (sums only, threshold, dropout error), not the cryptographic protocol itself.

5.5 Future work

The natural continuations, in roughly increasing order of scope: (i) referee-grade hardening of the corrected proposition - estimating β_{eff}, a formal Clarke-subdifferential treatment of the $\min(\cdot, \cdot)$ arm selection, and an explicit $O(1/T)$ constant; (ii) a defense that stops poisoning *without* discarding the rare-class signal - the filtering layer of Jiang et al. [12] is the closest precedent, and the defended-cell collapse of Section 4.8 is precisely the gap it would need to close; (iii) real cross-silo evaluation on NF-UQ-NIDS-v2, for which the four-silo partitioner and the bounded-memory loader already exist (T6 runs the T1 grid at 4 clients); (iv) verifiable SecAgg to remove the honest-execution assumption; and (v) the project's documented Option A - a few-shot/prototype-guided hybrid combining FedGRAA's gradient refinement with prototype exchange - which the remaining-compute plan identifies as a second, independent novelty contribution but which is deliberately not started pending a design/theory document.

Chapter 6

Conclusion

This internship set out to reproduce, diagnose, correct, and extend a federated-learning intrusion-detection paper, and to evaluate the result under the statistical and adversarial scrutiny the original publication did not receive. Each of the five objectives closed with a measured answer.

O1 - Faithful reproduction. The printed FedGRAA algorithm was implemented literally, with every documented ambiguity resolved by the code-level reading rather than patched over. The literal implementation reproduces none of the paper's headline numbers: 0.00% versus a claimed 88.88% rare-class accuracy on CIC-IDS-2017 at $\gamma = 1$, and 33.33% versus a claimed 21.09% on UNSW-NB15 - the latter an inversion in which the reproduction beats the claim while its loss diverges.

O2 - Diagnosis. The failure is a missing Euclidean projection onto the probability simplex. The printed update is an unconstrained descent over $\mathbb{R}^K$ while the paper's own lemma assumes $w \in \Delta$; the iterate leaves the simplex at round 1 in 120 of 120 unprojected cells, and the composite loss diverges by up to six orders of magnitude. The printed proof fails on three compounding grounds: convexity assumed for a non-convex tanh-MLP, no finite smoothness constant at the $\min(\cdot, \cdot)$ kink, and a tacit feasibility assumption nothing enforces.

O3 - Corrected theory. Projected subgradient descent $w^t = \Pi_\Delta(w^{t-1} - \eta' \partial L'(w^{t-1}))$ with $\eta' = 0.01$ restores feasibility structurally - 0 of 120 projected cells violate - and the corrected guarantee is gradient-mapping stationarity, $\min_{t \le T} \|G_{\eta'}(w^t)\|^2 = O(1/T)$, stated honestly as weaker than the paper's global-optimality claim. The Stage B factorial showed the fix is the only arm never catastrophically wrong on either metric, on either dataset, at any γ.

O4 - Privacy-compliant extension. FedGRAA-v2 removes the manuscript's two privacy assumptions - oracle rare-class knowledge and a server-held labelled validation set - via SecAgg-summed label histograms and a two-phase federated backward pass that is algebraically identical to the pooled computation ($\tilde{\zeta} \lesssim 4 \times 10^{-7}$). The ζ-inexact proposition generalizes the convergence result to $O(1/T) + O(\zeta^2)$.

O5 - Statistically tested evaluation. Across the 960-cell Stage C campaign, the best v2 arm ($m = 3$) reaches 93.47% rare-class recall on CIC-IDS-2017 and 69.66% on UNSW-NB15 - exceeding the privacy-violating oracle on both - at 3.00× FedAvg's communication cost, with 111 of 112 Holm-corrected comparisons significant. The adversarial campaigns quantify the mechanism's own attack surface (up to 3.3× uniform attacker weight) and document, at full precision, the finding that in every cell where an evaluated poisoning defense succeeds in stopping the attack, it does so only by destroying the rare-class signal the system exists to protect - 30/30 CIC defended cells and 19/30 UNSW defended cells collapse to exactly 0.0% rare recall.

The arc is therefore complete: the original mechanism works, once it is made to do what its own proof assumed it did. The honest version of the claim - a different, defensible point in an occupied design space, with an evaluation axis no adjacent work reports - is the one this report stands behind.

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Appendix A

Reproducibility Manifest

This appendix records the artifact-level details needed to reproduce or audit every number in this report. The project repository is `github.com/N1KH1LT0X1N/FedGRAA` (MIT license); the working tree is pinned at Python 3.13.14 with TensorFlow 2.21.0, Keras 3.15.0, NumPy 2.5.1, SciPy 1.18.0, scikit-learn 1.9.0 and pytest 9.1.1.

A.1 Artifact map

Every quantitative claim in this report traces to one of the generated artifacts or raw result files in Table A.1:

Table A.1: Figure and table provenance.

Report item	Source artifact(s)
Figs. 4.1a, 4.1b, 4.2a, 4.2b	`results/{DS}_gamma_sweep.json`, `{DS}_n_sweep.json`
Fig. 4.3	`results/diagnosis/`, `fedgraa_convergence_defect.md` §2
Figs. 4.4a, 4.4b; Tabs. 4.3, 4.4	`results/ablation/*_ablation_table.md` + raw JSON
Fig. 4.5a; Tab. 4.5	`results/v2_frontier/frontier_table.md` + raw JSON
Fig. 4.5b	`results/v2_frontier/` (classic arms)
Fig. 4.6; Tab. 4.7	`results/v2_frontier/overhead_table.md`
§4.6	`results/v2_frontier/significance_table.md` (112 rows)
Fig. 4.7; Tab. 4.8	`results/v2_frontier/*raresig*.json`, `attacker_weight_history`
Fig. 4.8; Tab. 4.9	`results/v2_frontier/{secfednids,weidetect}*.json`
Defended-cell counts (§4.8)	`final_last_n_acc` of the 60 defended JSON cells

A.2 Campaign accounting

The Stage C campaign comprises 960 result cells, each persisted as a self-describing JSON under `results/`: an 830-cell main grid (480 frontier/classic cells = 24 arms × 2 datasets × 10 seeds, plus 200 T2, 100 T4 and 50 T6 extension cells), a 90-cell T7 adversarial grid and a 40-cell T7G attacker-side γ sweep. Stage B comprises 240 diagnostic cells (4 arms × 2 datasets × 3 γ × 10 seeds) plus 40 baseline seed-runs; Stage A comprises the 200-round seed-0 γ and n sweeps.

A.3 Test suite

The repository's verification gates, wired in `.github/workflows/docs-check.yml`, are:

- `compileall` over `fedgraa`, `scripts` and `tests`; `pytest tests`; `check_doc_links.py`;
- `check_stage_a_complete.py` and `audit_stage_a.py` (Stage A);
- `check_stage_b_complete.py` and `audit_stage_b.py` (Stage B);
- `check_stage_c_complete.py` and `audit_stage_c.py` (Stage C; tiers `T1-T6` plus `TC` with `--t4-m 3`, and `T7` at seeds 0-4).

Test-count references across documents are snapshots, not a single invariant: 79 at the Stage A close, 142 at the Stage B close, 558 by `--collect-only` in the Stage C report's own recomputation, 563 in that report's measured suite, and **565** collected on the current working tree when the optional `matplotlib` and `modal` dependencies are present (their absence cleanly skips 35 collection-time-gated tests: 7 plot, 6 render-ablation, 22 modal). One script outside `tests/` is not part of the suite: the archived GPU probe `scripts/archive/gpu/test_gpu.py`, which fails on CPU-only TensorFlow builds by design (its CUDA build-info key is absent).

A.4 Re-derivation

Each stage ships an independent audit that recomputes its reported numbers from raw JSON rather than trusting the documents: `audit_stage_a`, `audit_stage_b` (SHA-256 manifest check, per-seed exact compare, SciPy recomputation of the 60 pairwise comparisons and 18 factorial effects, and 262 doc-transcribed numbers) and `audit_stage_c`. The exactness of the two-phase gradient protocol is enforced by `tests/test_aggregation_v2.py` (relative error $< 10^{-6}$) and measurable end-to-end via `scripts/verify_v2_exactness.py`; the byte-formula of Eq. (3.9) is enforced by `test_measured_bytes_match_frontier_formula`.